

Urinary System

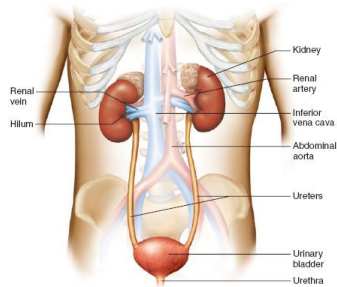


FIGURE 20.1 The urinary system includes the kidneys, ureters, urinary bladder, and urethra. Notice the relationship of these structures to the major blood vessels.

Functions of urinary system

- **Excretion** of many bioactive substances including many drugs
- Excretion of metabolic wastes along with excess water and electrolytes in urine
- **Regulation** of the **balance** b/n the **water** and **electrolytes** in the body and **acid-base** balance
- **Regulation** of arterial **blood pressure** by secretion of renin

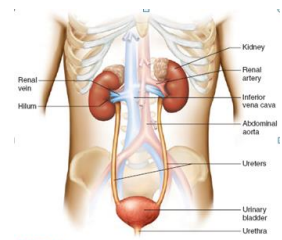
Cont...

Components of urinary system

- Kidneys
- Ureters
- Urinary bladder
- Urethra

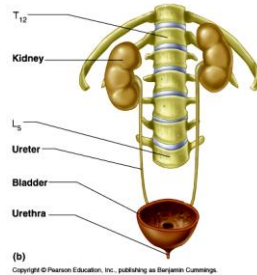
kidneys

- The red-brown, bean-shaped
- 10 cm in length, 5 cm in width, and 2.5 cm in thickness
- weights about 150 gm in males and 135gm in females
- lie **retroperitoneal** (behind the parietal peritoneum) in the **superior lumbar** region of the posterior abdominal wall



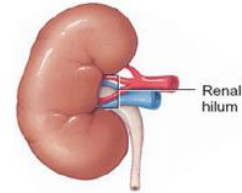
Cont....

- They are situated on each side of the **vertebral** column
- at the level of the **T12–L3** vertebrae
- The **right** kidney is usually slightly **inferior** to the left, reflecting its relationship to the **liver**



Cont....

- The lateral surface of each kidney is **convex**
- the medial surface is **concave** and has a vertical cleft called the **renal hilum**
- vessels, ureters, and nerves enter and leave the kidney through the renal hilum
- On the superior part of each kidney lies an **adrenal (suprarenal) gland** an endocrine gland



Internal anatomy of kidney

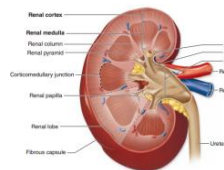
- The kidney has two distinct regions internally

1. Cortex

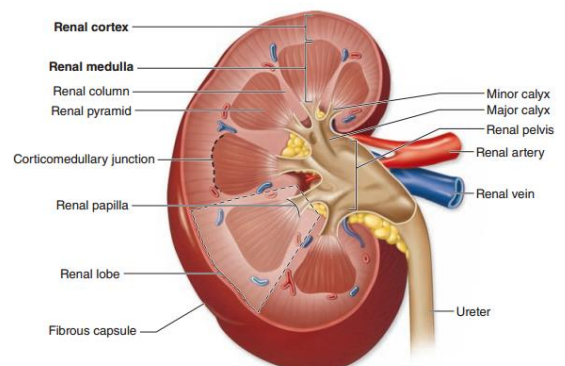
- Outer region
- consisting of many **corpuscles** and cross sections of **tubules**

2. Medulla

- Inner region
- **8-12** renal pyramids
- Separated by extensions from the cortex called **renal columns**



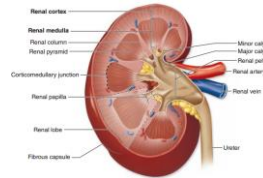
7



8

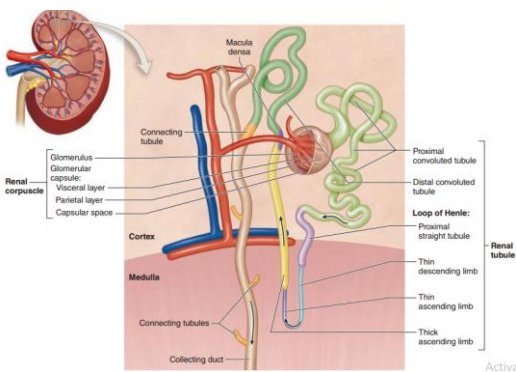
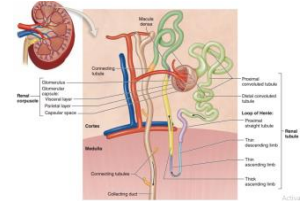
Cont....

- Each pyramid plus the cortical tissue at its base and along its sides constitutes a **renal lobe**
- Striations extending from the medulla into the cortex are called **medullary rays**; these plus the attached cortical tissue are considered **lobules**
- Tip of each pyramid, called the **renal papilla**, projects into a minor calyx that collects urine formed by **tubules** in the pyramid



Microscopic Anatomy of the Kidneys

- Each kidney contains around 2 million **nephrons**
- are the **structural** and **functional** units of kidney
- consist of simple, single layered epithelium along their entire lengths
- Are responsible for the production of **urine**
- consists of the renal **corpuscle** and a **tubule** system

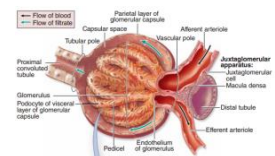


11

Cont....

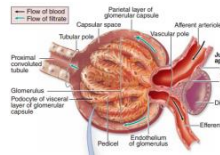
Renal corpuscle

- represents the beginning of the nephron
- Always located in the cortex
- consists of the:
 - ✓ **Glomerulus**
 - A tuft of capillaries composed of **10 to 20** capillary loops
 - Capillaries of each glomerulus have a total length of **1 cm**



Cont....

- This tuft of capillaries is **supplied** by an **afferent** arteriole and **drained** by an **efferent** arteriole
- The **endothelium** of the capillaries in the glomerulus is **fenestrated**

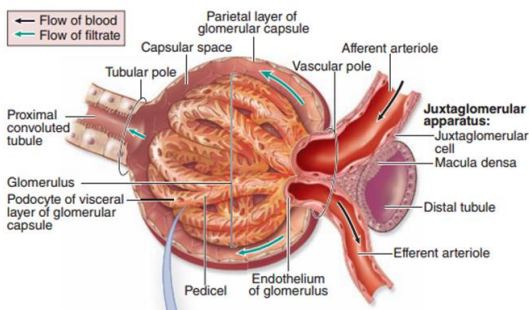
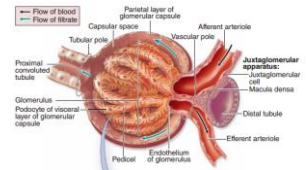


Cont....

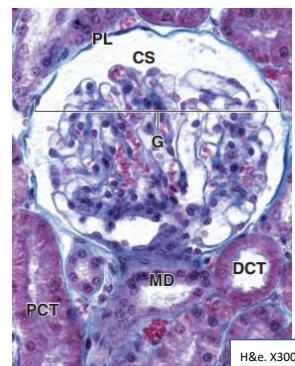
- thus these capillaries are highly **porous**, allowing large quantities of fluid and small molecules to pass from the capillary blood

✓ Bowman's capsule

- double-layered epithelial cup
 - Visceral layer
 - Parietal layer
- Between the two capsular layers is the **capsular (urinary) space**

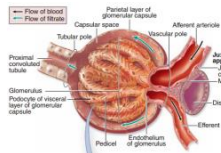


15



16

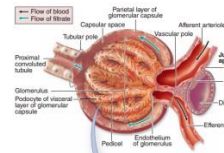
- Each renal corpuscle has:
- ✓ **vascular pole**
- where the afferent arteriole enters and the efferent arteriole leaves
- ✓ **tubular pole**
- where the proximal convoluted tubule begins



Cont....

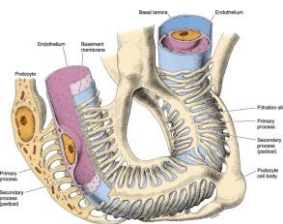
- The **parietal** layer of a glomerular capsule consists of a **simple squamous** epithelium

- At the tubular pole, this epithelium changes to the **simple cuboidal** epithelium that continues and forms the **proximal tubule**
- The parietal layer plays **no** part in the formation of filtrate



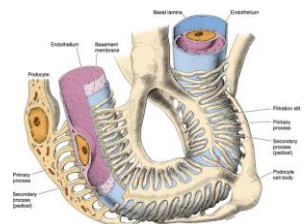
Cont....

- The visceral layer of a renal corpuscle consists of unusual stellate epithelial cells called **podocytes**
- From the cell body of each podocyte several **primary processes** extend and curve around a length of glomerular capillary



Cont....

- Each primary process gives rise to many parallel, interdigitating **secondary processes or pedicels**
- The pedicels cover much of the capillary surface, in direct contact with the basal lamina
- Between the interdigitating pedicels are elongated spaces, or **filtration slit pores**, 25 to 30 nm wide



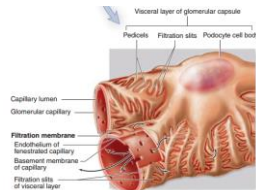
Cont....

Cont....

- filtration slits are covered by an ultrathin **filtration slit diaphragm** that spans the filtration slit slightly above the GBM

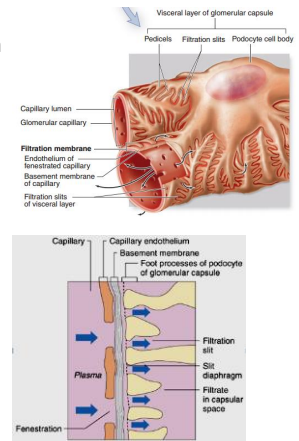
Filtration membrane

- is the actual filter that lies between the **blood** and the interior of the **glomerular capsule**



- Filtration occurs through a structure with three parts:

- fenestrations of the **capillary endothelium**
 - possesses numerous fenestrations (70 to 90 nm in diameter)
 - blocks **blood cells** and **platelets**
- Basement membranes**
 - 300 to 370 nm thick
 - restricts **large proteins** and some **organic anions**

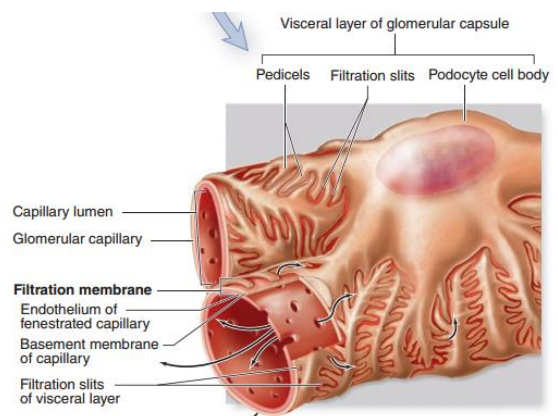
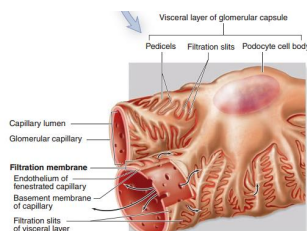


Cont....

3. Filtration slit diaphragms

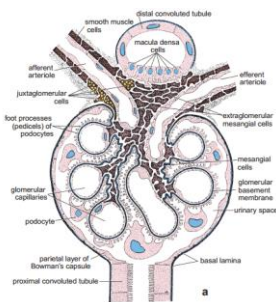
- between **pedicels**
- restrict some small proteins and organic anions

- The filtrate passes into the capsular space
- about 20% of the blood plasma entering a glomerulus is filtered into the capsular space



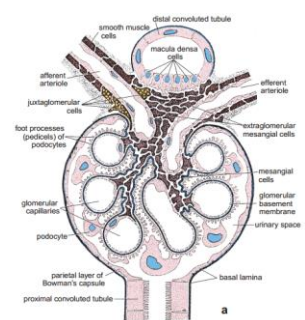
Cont....

- The renal corpuscle also contains group of cells called **mesangial cells**
- These cells and their extracellular matrix constitute the **mesangium**
- It is most obvious at the vascular stalk of the glomerulus and at the interstices of adjoining glomerular capillaries



Cont....

- The mesangial cells are not confined entirely to the renal corpuscle;
- some are located outside the corpuscle along the vascular pole,
- where they are also designated as **lacis cells** and form part of **juxtaglomerular apparatus**



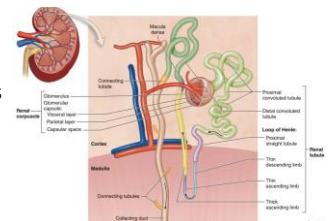
Cont...

- Functions of the mesangium :
 - **Physical support** of **capillaries** within the glomerulus
 - Adjusted contractions in response to blood pressure changes, which help **maintain** an optimal **filtration rate**
 - **Phagocytosis** of protein aggregates adhering to the glomerular filter, including antibody-antigen complexes abundant in many **pathological conditions**
 - **Secretion** of several cytokines, prostaglandins, and other factors important for immune **defense** and **repair** in the glomerulus

Nephron tubule parts

✓ Proximal tubule

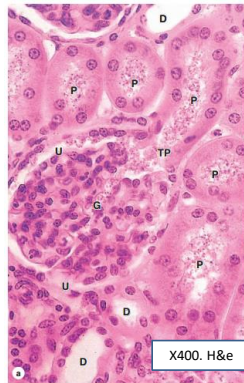
- a long convoluted part
- Located entirely in the cortex, with a shorter straight part that enters the medulla
- simple cuboidal epithelium



Cont....

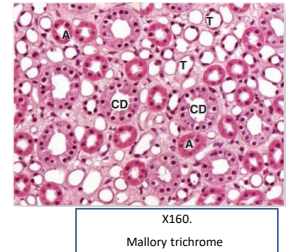
- cell apex has very many long microvilli that form a prominent brush border in the lumen that facilitates reabsorption
- ✓ **Loop of Henle (or nephron loop)**
- U-shaped structure
- with a thin descending limb and a thin ascending limb

Cont....



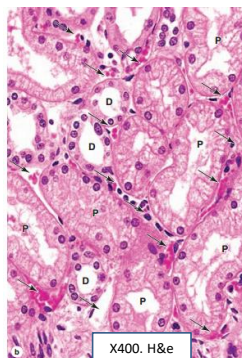
- both composed of **simple squamous** epithelia
- The thin ascending limb of the loop becomes the **thick ascending limb**
- with **simple cuboidal** epithelium again, in the outer medulla and extends as far as the **macula densa** near the nephron's glomerulus

Cont....

✓ **Distal tubule**

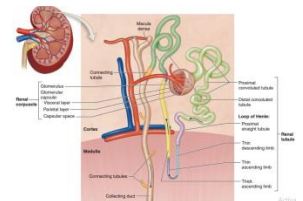
- consisting of a thick straight part ascending from the loop of Henle back into the cortex
- convoluted part completely in the cortex
- simple cuboidal cells having no brush border and more empty lumens

Cont....

✓ **Connecting tubule**

- short final part
- linking the nephron to collecting ducts
- Connecting tubules from several nephrons merge to form collecting tubules that then merge as larger collecting ducts
- These converge in the renal papilla, where they deliver urine to a minor calyx

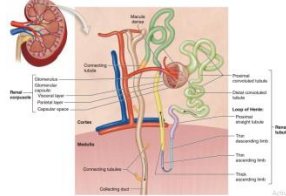
Cont....



Cont....

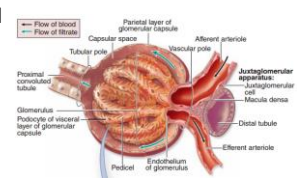
Collecting Ducts

- **connecting tubule** extends from each nephron and several join together to form collecting ducts
- **Simple cuboidal epithelium**
- an average diameter of **40 μm**
- In the apex of the pyramid, several collecting ducts merge further as a **papillary duct** which delivers urine to the **minor calyx**



Juxtaglomerular Apparatus

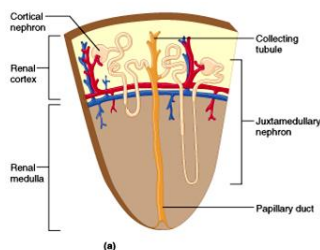
- includes:
 - macula densa
 - Juxtaglomerular cells
 - Extraglomerular mesangial cells
- regulate glomerular blood flow and keep the rate of glomerular filtration relatively constant



Cont....

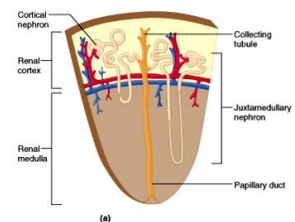
Classes of nephron

- **Cortical nephrons**
 - located almost entirely with in the cortex
 - They have short loops of Henle, extending only into the outer medulla
- **Juxtamedullary nephrons**
 - near the medulla



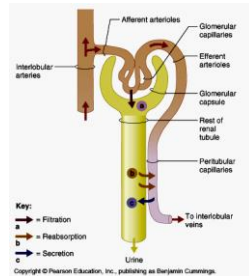
- their renal corpuscles lie near the cortex-medulla junction
- have long loops of Henle
- **Intermediate nephrons or midcortical nephrons**
 - Have their renal corpuscles in the midregion of the cortex
 - Their loops of Henle are of intermediate length

Cont....



Mechanisms of Urine Production

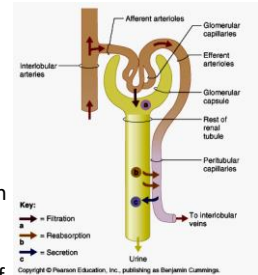
- The **uriniferous** tubule produces urine through three interacting mechanisms
- Filtration
- Reabsorption
- Secretion



37

Cont...

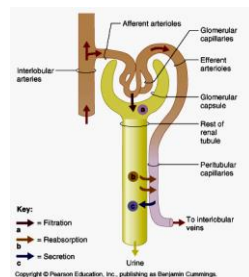
- In filtration a filtrate of the blood leaves the kidney capillaries and enters the nephron
- This filtrate resembles tissue fluid in that it contains all the **small** molecules of **blood** plasma
- As filtrate proceeds through the uriniferous tubule, the filtrate is processed into **urine** by the mechanisms of **reabsorption** and **secretion**



38

Cont...

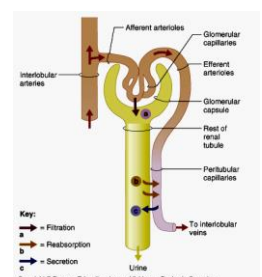
- During reabsorption, most of the **nutrients**, **water**, and essential ions are **reclaimed** from the filtrate, and returned to the blood of capillaries in the surrounding connective tissue
- 99% of the volume of renal filtrate is **reabsorbed**



39

Cont...

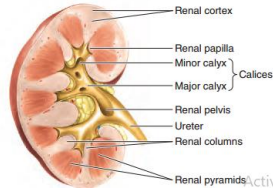
- As the **essential molecules** are reclaimed from the filtrate, the remaining wastes and needed substances contribute to the **urine** that ultimately leaves the body
- Secretion** is an active process which moves **undesirable** molecules into the tubule from the blood of surrounding **capillaries**



40

Cont...

- Urine flows from renal papilla → Several minor calyces → 2-3 major calyces → renal pelvis → ureter → urinary bladder
- The walls of the calyces, pelvis, and ureter contain **smooth muscle** which contract to move urine.



41

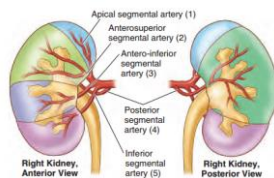
Blood Supply

- The kidney continuously **cleanse** the blood and adjust its composition
- Kidneys possess an extensive blood supply
- Under normal resting conditions, the renal arteries deliver approximately **one-fourth** of the total systemic cardiac output (1200 ml) to the kidneys each minute

42

Cont...

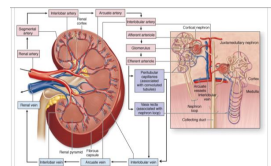
- The renal arteries arise at right angles from the abdominal aorta
- Each renal artery divides into five **segmental** arteries that enter the hilum
 - Superior segmental artery
 - anterosuperior segmental
 - antero-inferior segmental
 - inferior segmental artery
 - posterior segmental artery



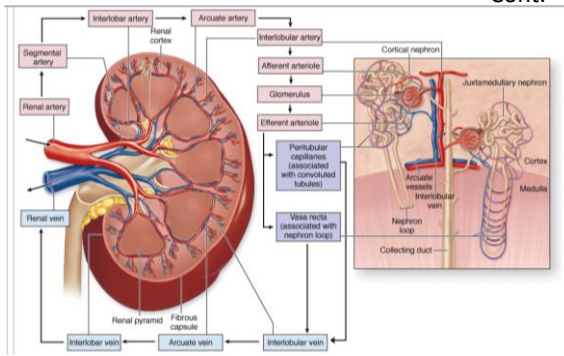
43

Cont....

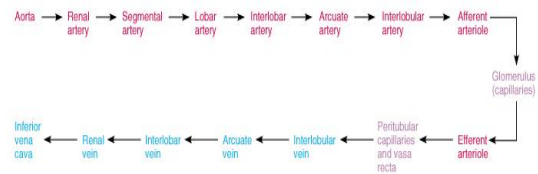
- Each segmental artery divides into **lobar** and **interlobular** arteries
- Several **renal veins** drain each kidney and unite in a to form the **right and left renal veins**



Cont.



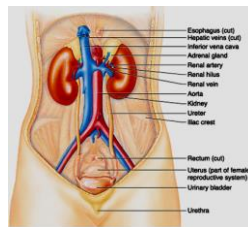
Blood circulation



Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

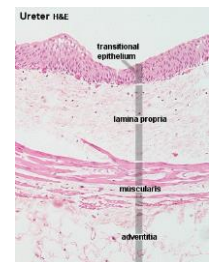
Ureters

- The ureters are slender tubes
- **convey** urine from the kidneys to the bladder
- Each leaves the renal pelvis, descends behind the **peritoneum** to the base of the bladder,
- turns and then runs obliquely through the **medial** bladder wall



Cont....

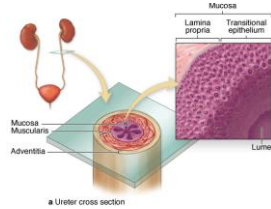
- Histologically, the walls of the ureter is tri layered
1. An inner layer of **transitional** epithelium lines the inner mucosa
 2. The middle **muscularis** layer is composed of an inner **longitudinal** layer and an outer **circular** layer
 3. The outer layer is composed of fibrous **connective** tissue



Cont...

- Cells of this epithelium are organized as three layers:

1. A **single** layer of small **basal** cells resting on a very thin **basement membrane**
2. An **intermediate region** containing from one to several layers of more columnar cells

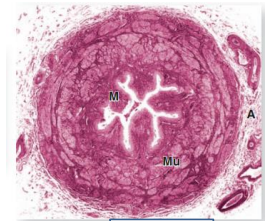


49

3. A **superficial** layer of very large, bulbous cells called **umbrella** cells

- are occasionally **bi- or multinucleated** and are highly differentiated to protect underlying cells against the **cytotoxic** effects of hypertonic urine

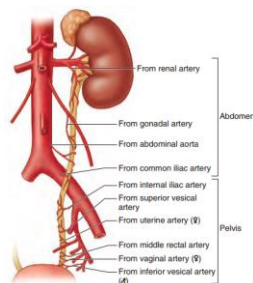
Cont....



X18. H&e

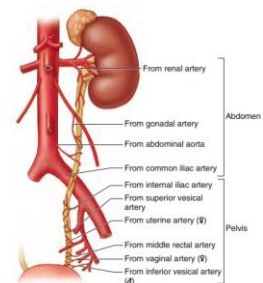
Cont....

- arterial supply
- Abdominal portion**
- renal arteries
- testicular or ovarian arteries
- abdominal aorta
- common iliac arteries
- Pelvic Portion**
- common iliac
- internal iliac
- ovarian arteries



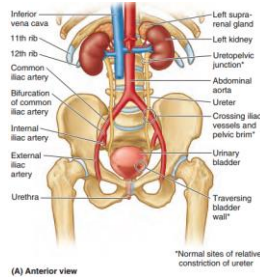
Cont....

- uterine arteries
- inferior vesical arteries
- ✓ **venous** drainage of the ureters generally **parallels** the arterial supply



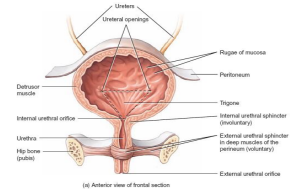
Urinary bladder

- temporary **reservoir** for urine
- Empty adult urinary bladder is located in the **lesser pelvis**
- infants and young children, the urinary bladder is in the **abdomen** even when empty



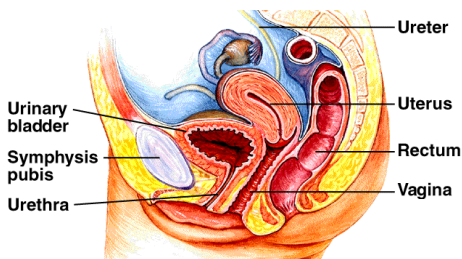
Cont....

- It can expand for storage or collapse when empty
- Empty its walls are thick and thrown into folds (**rugae**)
- As it expands it becomes pear shaped and rises in the **abdominal cavity**
- The bladder can store more than 300 ml of urine without a significant increase in **internal pressure**



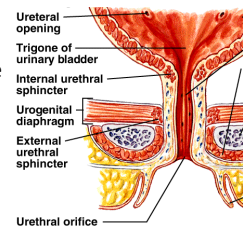
Cont...

- A **moderately** full bladder holds approximately 500 ml and can about 1000 ml at **capacity**
- In females, the bladder is **anterior** to the **vagina** and **uterus**



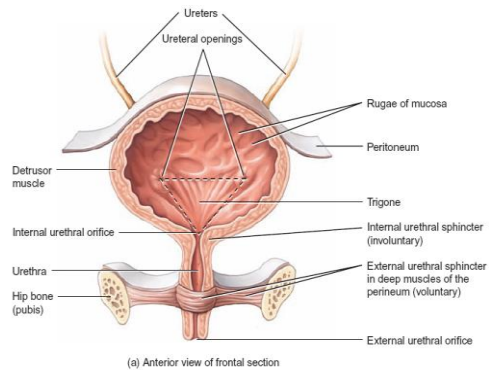
Cont.....

- The interior of the bladder has openings for both **ureters** and the **urethra**
- The triangular region of the bladder base outlined by these openings is called the **trigone** which is a common site of infections



55

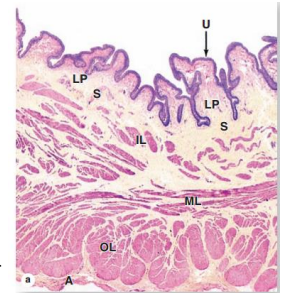
56



57

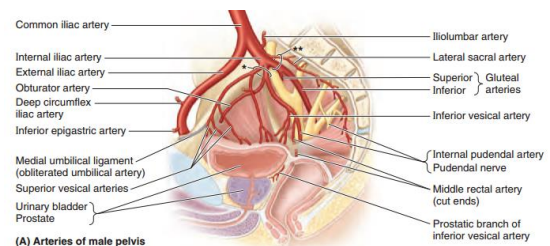
Histology of Urinary Bladder

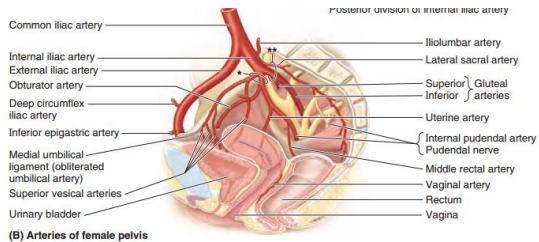
- The bladder wall has three layers
 - A **mucosa** containing transitional epithelium
 - A thick **muscular** layers
 - A fibrous **adventitia**
- The muscle layer consists of smooth muscle arranged inner and outer longitudinal layers
- Collectively the muscle layer is called the **detrusor muscle**



Cont....

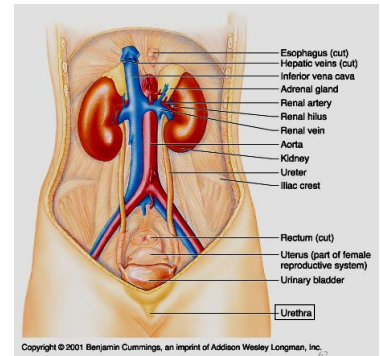
- The main arteries supplying the bladder are branches of the **internal iliac** arteries
- **superior vesical arteries**
- **inferior vesical arteries**
- **vaginal arteries**
- **Obturator** arteries
- **inferior gluteal** arteries
- The veins draining blood from the bladder correspond to the arteries





Urethra

- The *urethra* is a thin muscular tube that **drains urine** from the bladder and conveys it out of the body

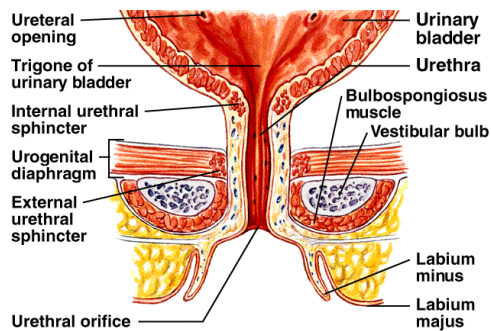


Histology of Urethra

Cont...

- The epithelium of its mucosal lining is mostly **Pseudostratified columnar** epithelium
- Near the bladder it is **transitional** epithelium and near its external opening it changes to a protective **squamous** epithelium

- At the bladder-urethra junction a thickening of the **detrusor** muscle forms the **internal sphincter** is **involuntary** sphincter that keeps the urethra closed when urine is not being passed
- A second sphincter, the **external urethral sphincter**, surrounds the urethra and is composed of skeletal muscle and thus is under **voluntary** control



Cont....

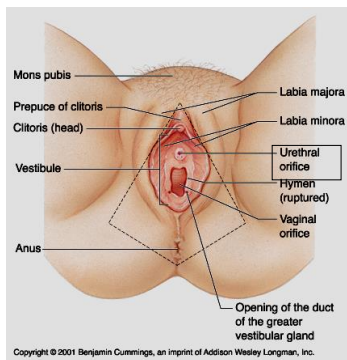
- The **levator ani** muscle of the pelvic floor also serves as a **voluntary constrictor** of the urethra
- The length and functions of the urethra differ in the two sexes
- In females the urethra is **3-4 cm** long and is tightly bound to the **anterior vaginal wall** by fibrous connective tissue

65

66

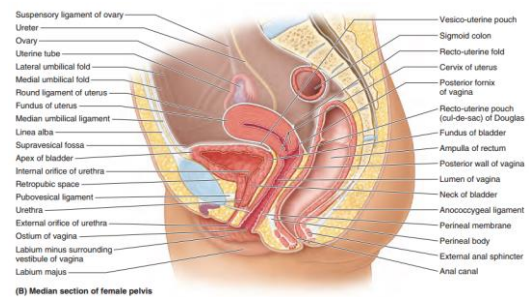
Cont....

- Its external opening, the external urethral orifice, **anterior** to the vaginal opening and **posterior** to the clitoris



Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

67

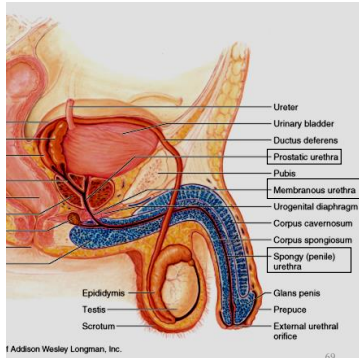


(B) Median section of female pelvis

68

Urethra

- In males the urethra is 20 cm long with three regions
 - a. Prostatic urethra
 - b. Membranous urethra
 - c. Spongy or penile urethra



Cont...

- The male urethra has two basic functions
 - ✓ It carries urine out of the body
 - ✓ It carries semen into the female reproductive tract

